Waqas Ahmad

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Nationality: Pakistani | **Gender:** Male | **Phone number:** (+92) 03369832077 (Mobile) |

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ABOUT ME

AI focused Computer Science undergraduate with research experience in healthcare deep learning and explainable models. Aspiring to advance medical AI innovation through postgraduate research.

EDUCATION AND TRAINING

01/10/2022 – CURRENT Mardan, Pakistan

BACHELOR OF SCIENCE IN COMPUTER SCIENCE Abdul Wali Khan University Mardan

Pursuing a Bachelor's in Computer Science with research focused on applying Machine Learning to Autism Spectrum Disorder (ASD). Authored multiple manuscripts including two published peer-reviewed article and several systematic reviews and articles currently under review on AI in ASD. Currently leading a project on multimodal fusion of fMRI, behavioral, and demographic data for explainable and fairness-aware ASD detection. My work emphasizes subgroup fairness, interpretability, and clinical applicability in computational neuroimaging. Dedicated to advancing AI-driven healthcare solutions through rigorous and impactful research. Maintained a CGPA of 3.0/4.0 till the 7th semester and committed to contributing to impactful research at the intersection of technology and healthcare

Website www.awkum.edu.pk

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C2	C2	C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Technical Skills

Python | Matplotlib | Keras | Java (computer programming) | Tensor-Flow | Pandas | C++ | Nilearn | Python (pandas, numpy, matplotlib) | PyTorch | Statistical Modelling | OpenCV | Scikit-Learn | Seaborn | Numpy | fMRI Preprocessing

Research & Analytical Skills

Research Skill | Overleaf & LaTeX | Literature Review & Gap Identification | Systematic Literature Review | Critical Thinking. | Critical Thinking and Analytical Skills | Problem-Solving Skills

Soft Skills

Academic Writing and Publication Preparation | Team Collaboration | Time Management | Critical Thinking

PUBLICATIONS

2025

Reproducible Machine Learning QSAR Pipeline for JAK2 Inhibitors with Structural Docking Validation

This work presents a reproducible ML-QSAR pipeline for predicting JAK2 inhibitors, using cheminformatics filtering, ECFP4/RDKit features, and SVR (R^2 = 0.71, Pearson = 0.85). Top predictions were validated by docking, showing binding modes and affinities comparable to Baricitinib. The entire pipeline data, code, environment is openly shared, promoting FAIR, reusable drug discovery workflows.

Authors: Waqas Ahmad, Hashim Ali, Aftab Ahmed, Umer Tanveer, Shaista Bibi | **Journal Name**: Annual Methodological Archive Research Review | **Volume, Issue and Pages**: Volume3,Issue12(2025)

2025

SCAFFOLD-AWARE MACHINE LEARNING–DOCKINGPIPELINE FOR TYK2 INHIBITOR DISCOVERY WITHCALIBRATED PRIORITIZATION OF32 ACTIVES INCLUDING DEUCRAVACITINIB

This paper presents a scaffold-aware machine learning–docking pipeline for TYK2 inhibitor discovery, integrating rigorous data curation, calibrated XGBoost classification (ACC = 0.875, AUC = 0.951), and physics-based validation to prioritize 32 high-confidence actives including the FDA-approved drug Deucravacitinib. The approach ensures generalizability to novel chemotypes via scaffold-split validation and provides a reproducible, open framework for kinase-targeted drug repurposing.

Authors: Waqas Ahmad, Hashim Ali, Aftab Ahmed, Umer Tanveer, Shaista Bibi | **Journal Name**: Spectrum of Engineering science | **Volume, Issue and Pages**: Volume 3, Issue 12, 2025

2025

Calibrated Ensemble Learning for Autism Spectrum Disorder Classification Using Multi-modal ABIDE Data

Title: Calibrated Ensemble Learning for Autism Spectrum Disorder Clas si cation Using Multi-modal ABIDE Data

- **Status:** Under Review
- **Role:** First Author
- **Journal/Conference Name:** Journal of Computational Biology and Medicine
- **Publisher:** ScienceDirect
- **Description:**

A calibrated ensemble framework that integrates multimodal neuroimaging and behavioral data to enhance reliable and interpretable Autism Spectrum Disorder classification.

Authors: WAQAS AHMAD, ASHRAF ZIA | **Journal Name**: Journal of Computational Biology and Medicine | **Volume, Issue and Pages**: Under Review | **Publisher**: ScienceDirect

2025

The Triple Challenge: A Systematic Review of Accuracy, Explainability, and Fairness in Artificial Intelligence for Autism Spectrum Disorder

Title: The Triple Challenge: A Systematic Review of Accuracy, Explainability, and Fairness in Artificial Intelligence for Autism Spectrum Disorder
• Status: Under Review
• Role: First Author
• Journal/Conference Name: Journal of Autism and Developmental Disorders
• Publisher: IEEE
• Description: A systematic review of AI methods for ASD detection, addressing the triple challenges of accuracy, explainability, and fairness. The article identifies key gaps in current research and outlines pathways toward trustworthy, clinically applicable AI systems.

Authors: WAQAS AHMAD, ASHRAF ZIA | **Journal Name**: Journal of Autism and Developmental Disorders | **Publisher**: Springer

2025

DevOps, Agile, and Lean: A Systematic Literature Review of Integration Strategies

Title: DevOps, Agile, and Lean: A Systematic Literature Review of Integration Strategies

- **Status:** Under Review
- **Role:** First Author
- **Journal/Conference Name:** Discover Computing
- **Publisher:** Springer

Description:

A systematic review analyzing how **DevOps, Agile, and Lean methodologies** interact in software engineering. The study synthesizes integration strategies, success factors, and challenges to guide practitioners in improving software delivery and project outcomes.

Authors: WAQAS AHMAD, ASHRAF ZIA | **Journal Name**: Discover Computing | **Publisher**: Springer Nature

2025

Optimization of Multivariate Pre-Processing Techniques for Analyzing Near Infra-Red Spectroscopic Data and Partial Least Square Predictive Mode

Title: Optimization of Multivariate Pre-Processing Techniques for Analyzing Near Infra-Red Spectroscopic Data and Partial Least Square Predictive Mode
Status: Under Review
Role: Co-author
Journal/Conference Name: IEEE Access
Publisher: Springer
Description:
This study evaluates and optimizes **multivariate preprocessing techniques** for NIR spectroscopic data to enhance the performance of **partial least squares predictive modeling**, improving accuracy in chemometric analysis.

Authors: MALAIKA WAHAB, ASHRAF ZIA, WAQAS AHMAD | **Journal Name**: IEEE Access | **Publisher**: IEEE

2025

Student Performance Evaluation with Artificial Intelligence and Machine Learning Through Data Mining

Title: Student Performance Evaluation with Artificial Intelligence and Machine Learning Through Data Mining

- **Status:** Under Review
- **Role:** Co-Author
- **Journal/Conference Name:** IEEE Access
- **Publisher:** IEEE

Description:
This work applies **AI and machine learning–based data mining techniques** to predict and evaluate student academic performance, aiming to support decision-making in **educational analytics and early intervention**.

Authors: UZAIR SAID, MUHAMMAD SHAHAB ZADA, OSAMA ALI SHAH | **Journal Name**: IEEE Access | **Publisher**: IEEE

2025

From Learning to Action: Harnessing AI, ML, and DL for Next-Generation Robotics Detection

Description: This review explores the integration of AI, machine learning, and deep learning in robotics for complex environments. It emphasizes advancements in perception, decision-making, and adaptability, while identifying challenges for real-world multi-sector applications.

Authors: WAQAS AHMAD, ASHRAF ZIA | **Journal Name**: Journal of Intelligent & Robotic Systems | **Publisher**: Springer

HOBBIES AND INTERESTS

Coding

Enjoy developing small projects, exploring new algorithms, and building a deeper understanding of software systems and machine learning models.

Reading

Passionate about reading research articles, technical books, and literature related to software engineering, machine learning, and emerging technologies.

Running

Enjoy running as a way to maintain physical fitness, build discipline, and refresh the mind for better focus and productivity.

CERTIFICATIONS

Introduction to Artificial Intelligence (AI)

- Investigated various AI techniques including machine learning, deep learning, and natural language processing.
- Explored the fundamental concepts of Artificial Intelligence and its historical evolution.
- Collaborated on a capstone project that applied AI methods to a significant challenge in healthcare.

Python for Data Science, AI & Development

- Developed multiple machine learning models using Python for predictive analysis in healthcare applications.
- Conducted in-depth analyses of software development methodologies utilizing Python for automation of DevOps pipelines.
- Implemented Explainable AI techniques in research projects to enhance the interpretability of data-driven decisions.

Introduction to Software Engineering

- Developed a comprehensive understanding of software development methodologies including Agile, Waterfall, and DevOps.
- Applied software engineering principles to analyze and design complex systems in practical projects.
- Conducted in-depth research on the integration of Agile and DevOps methodologies to enhance software delivery processes.

Introduction to Agile Development and Scrum

- Developed a comprehensive understanding of Agile methodologies and their application in software development.
- Explored the principles and practices of Scrum as a framework for managing software projects.
- Participated in collaborative team exercises to simulate Agile project workflows and enhance teamwork skills.

ACHIEVEMENTS

Academic and Research Accomplishments

- Published two peer-reviewed articles in well reputable journals
- Authored three first-author systematic reviews at the intersection of AI and domain-specific challenges: (1) AI applications in Autism Spectrum Disorder (ASD), (2) AI/ML in robotics systems, and (3) AI-enabled optimization in DevOps, Agile, and Lean software development. Additionally, co-authored two studies on spectroscopic data preprocessing and AI-driven evaluation of student performance.
- Currently leading a technical research project on multimodal fusion of fMRI and behavioral data for explainable and fair Autism Spectrum Disorder detection.
- Maintained a CGPA of 3.0/4.0 till 7th semester.
- Specialized in research focused on Software Engineering and Machine Learning, demonstrating a strong foundational knowledge in these areas.
- Developed strong technical, research, and analytical skills through academic coursework and independent projects.
- Committed to contributing to impactful research at the intersection of technology and healthcare.
- Contributed to advancing knowledge in technology and healthcare through rigorous research methodologies and peer-reviewed publications.